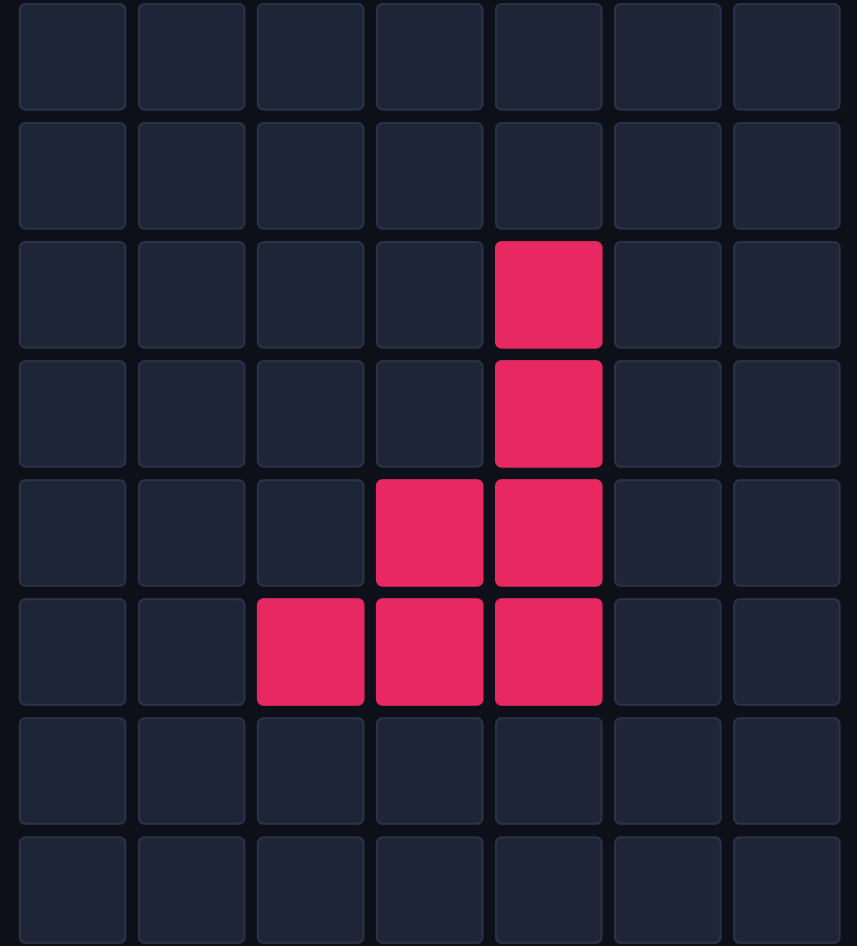


Perception-Aware Dynamic Diffusion Transformers

for Safety-Critical Scene Generation

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MOTIVATION

The rarest frames are the ones that matter most

The data problem

An autonomous stack sees millions of ordinary frames and almost none of the ones that decide whether someone lives. A pedestrian in dark clothing under a streetlight is rare in any collected dataset, expensive to capture deliberately, and unsafe to capture at the frequency training needs.

Generation is the standard answer

Layout-conditioned diffusion lets you author a scene a cyclist here, a pedestrian there and synthesise frames on demand. The tail becomes something you specify rather than something you wait for.

capacity is finite, so something must choose

A diffusion transformer cannot spend equal compute on every region of every frame. Some allocator decides which parts get fine-grained capacity. That allocator is the subject of this project.



THE PROBLEM

Entropy is not importance

WHAT ENTROPY SEES

A textured tree line, cobblestones, building facades. High pixel variance, zero relevance to braking.

WHAT ACTUALLY MATTERS

A pedestrian in dark clothing. Smooth, low-contrast, low entropy and the one object a car must not hit.

scene



entropy spends here



ours spends here

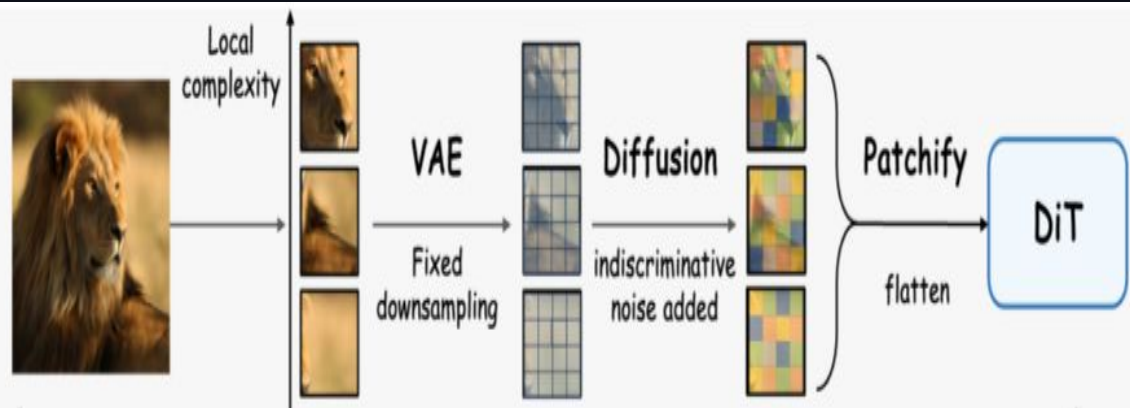


Every existing allocator asks the pixels, not the scene

Work	Contribution	Allocates on
DiT (Peebles & Xie)	Transformer over latent patches; the adaLN-Zero conditioning we inherit	uniform patches
D ² iT ← direct parent	Dynamic grain: per-region capacity, fine tokens where needed	Pixel entropy

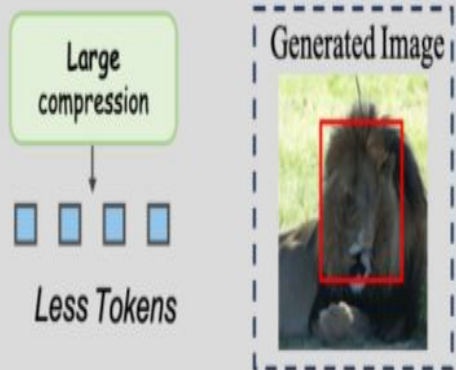
Layout information already exists in every one of these systems. It conditions the output. It has never been used to allocate the compute.

RELATED WORK



(i) fixed **large** compression:

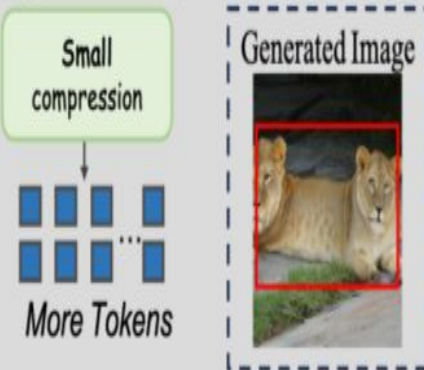
Limited local realism



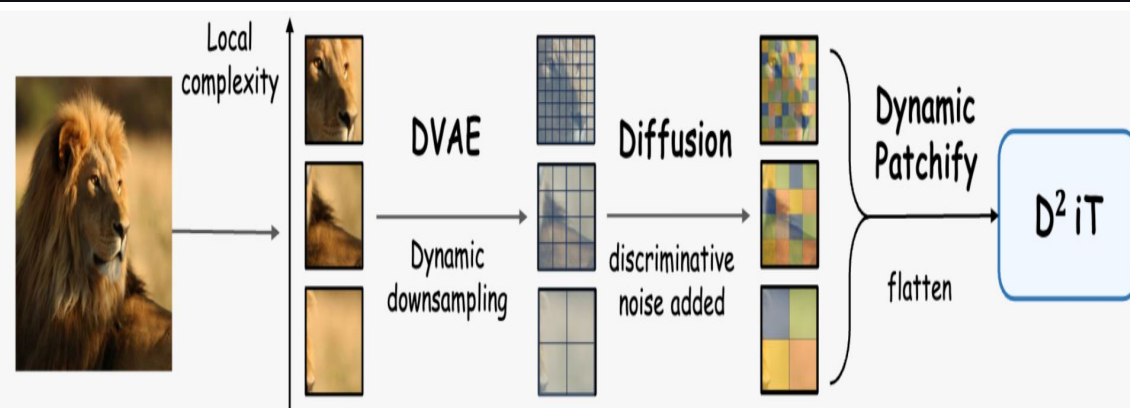
or

(ii) fixed **small** compression:

Limited global consistency & High computational complexity



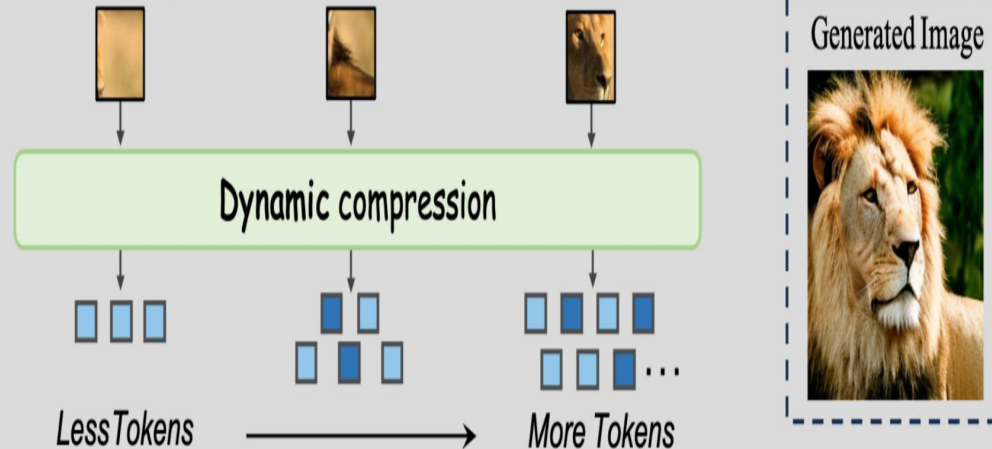
DiT (Peebles & Xie)



dynamic compression: Unified global consistency & local realism

Smooth regions

Detail regions

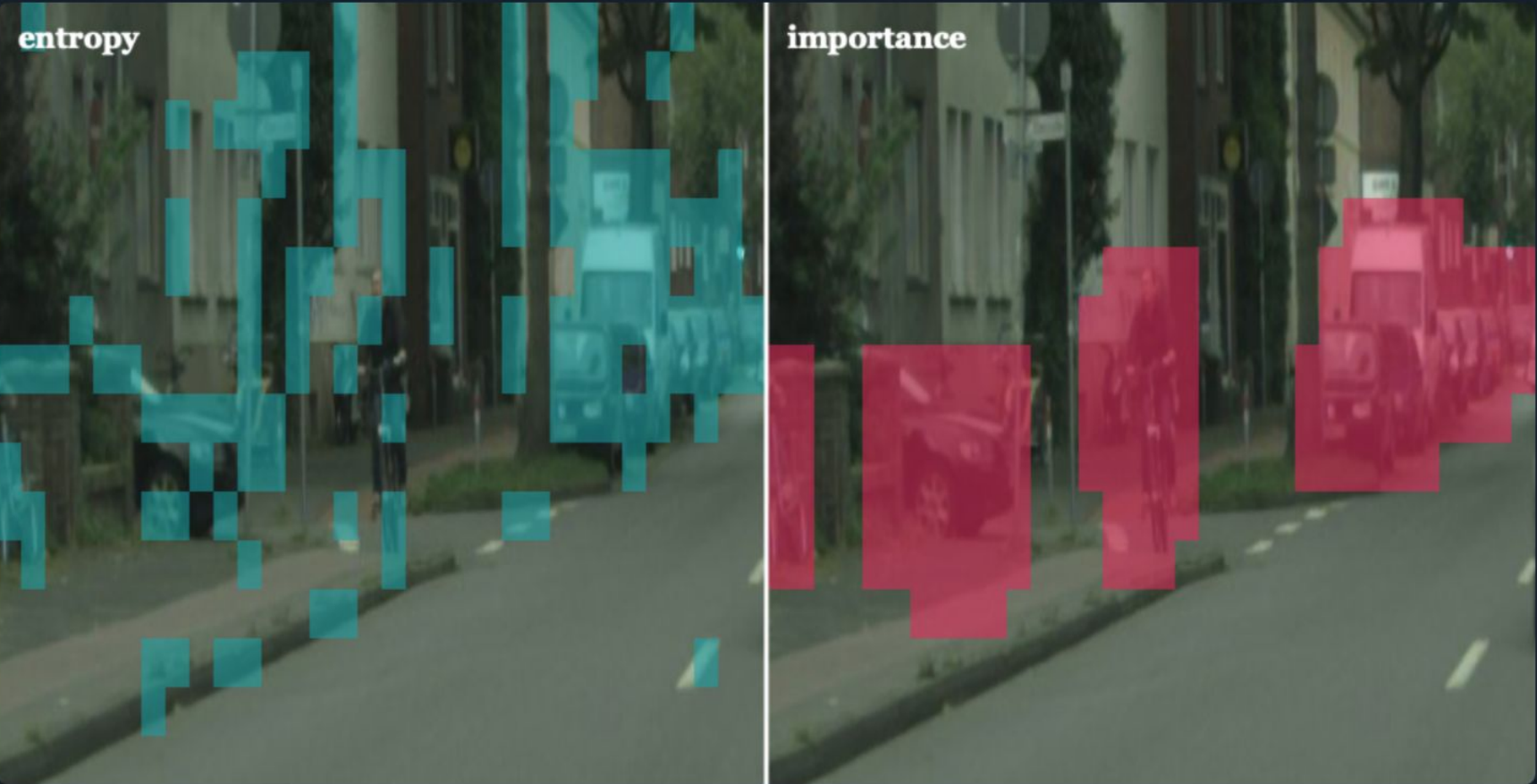


D²iT ← direct parent

Our Work: Perception-Aware Diffusion

512 cells per image, only 128 get high-detail tokens. Every strategy spends the **same 128 cells, 896 tokens, 26.9 GFLOPs** — only the placement differs.

- importance
- entropy
- random
- block
- scatter



- WIN recall +78 pts
- WIN background -0.0021
- WIN VRU -0.0216

strategy	importance
seams	88 (ent 253)
separate patches	4
VRU cells reached	100% (ent 22.2%)
pedestrians	1
VRU error here	0.7449
bg error here	0.5979

Layout-derived. Fine cells land on people in contiguous blobs — few seams, high recall.

- ☒ outline pedestrians
- ☒ cell grid
- ☒ compare with entropy

What we ran it on, and what we measured

DATASET

- **Dataset** - Cityscapes fine annotations, official scene-level split.
- 2,975 train / 500 val frames (85.6% / 14.4%), no scene overlap.
- Cropped to 256×512: 2 crops per train frame, 1 per val frame → **5,950 train / 500 val crops (6,450 total)**.
- Horizontal-flip augmentation gives 11,900 effective training samples.
- No test split Cityscapes test labels are withheld on the benchmark server, so evaluation is on val, as is standard.

MODELS

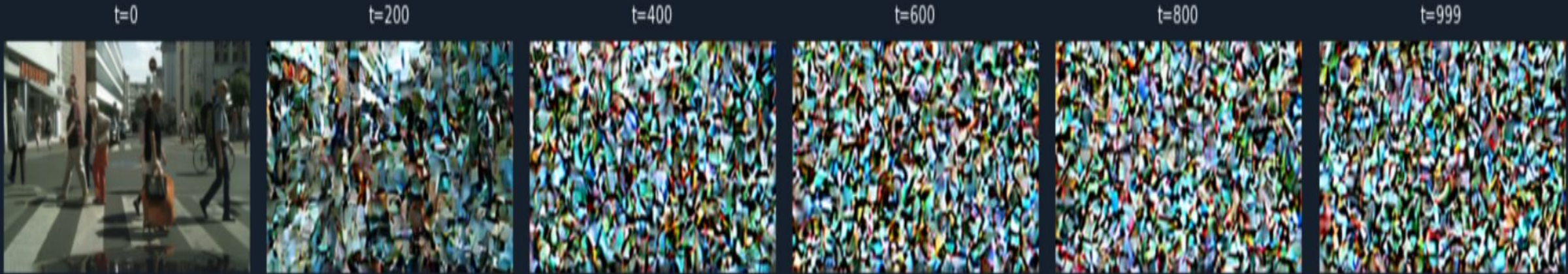
- **SmallIDiT** 7 M params, width 192, depth 8, 6 heads, uniform 2×2 patchify, layout cross-attention, adaLN-Zero
- **StableDiffusion v1.5** (runwayml/stable-diffusion-v1-5) + **LoRA** rank 16, 4.4 M trainable against 863 M frozen UNet
- Zero-initialised layout adapter on the cross-attention context
- Frozen throughout: sd-vae-ft-mse VAE (f = 8) and CLIP ViT-L/14 text encoder
- DiT-XL/2 saw 1.28 M distinct images. We have 2,975 training scenes, a ~430× deficit

METRICS

- **Allocation:** precision, recall, F1 against the VRU ground-truth mask reported against the attainable ceiling of 56%, never 1.0
- **Baselines:** analytic 25% chance line; unweighted arm as control
- **Controls:** fixed 25% cell budget, seed-locked identical batches, matched compute across arms

METHOD

How diffusion Models Work



NN Tries to fully predict noise at each time steps. Realistically, its just a prediction. we need multiple steps to get high quality images.

positions receiving the safety weight



Pipeline of Diffusion Model

Forward process — training

Gradually destroy an image by adding Gaussian noise

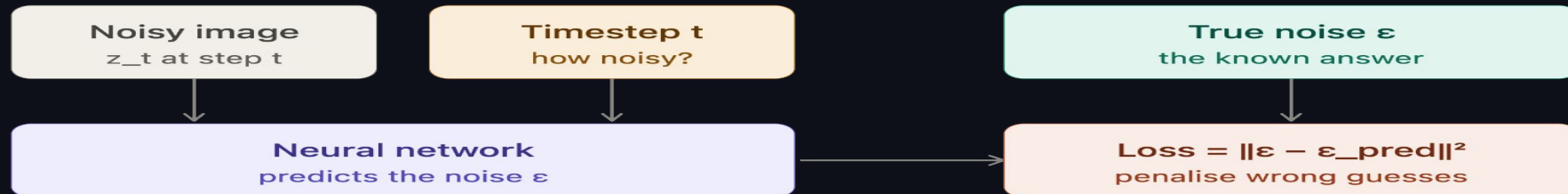


$$z_t = \sqrt{\alpha_t} \cdot z_0 + \sqrt{(1 - \alpha_t)} \cdot \epsilon$$

signal fades as noise grows — controlled by one number, the timestep t

Training objective — learn to predict the noise

Given a noisy image and its timestep, predict the noise that was added



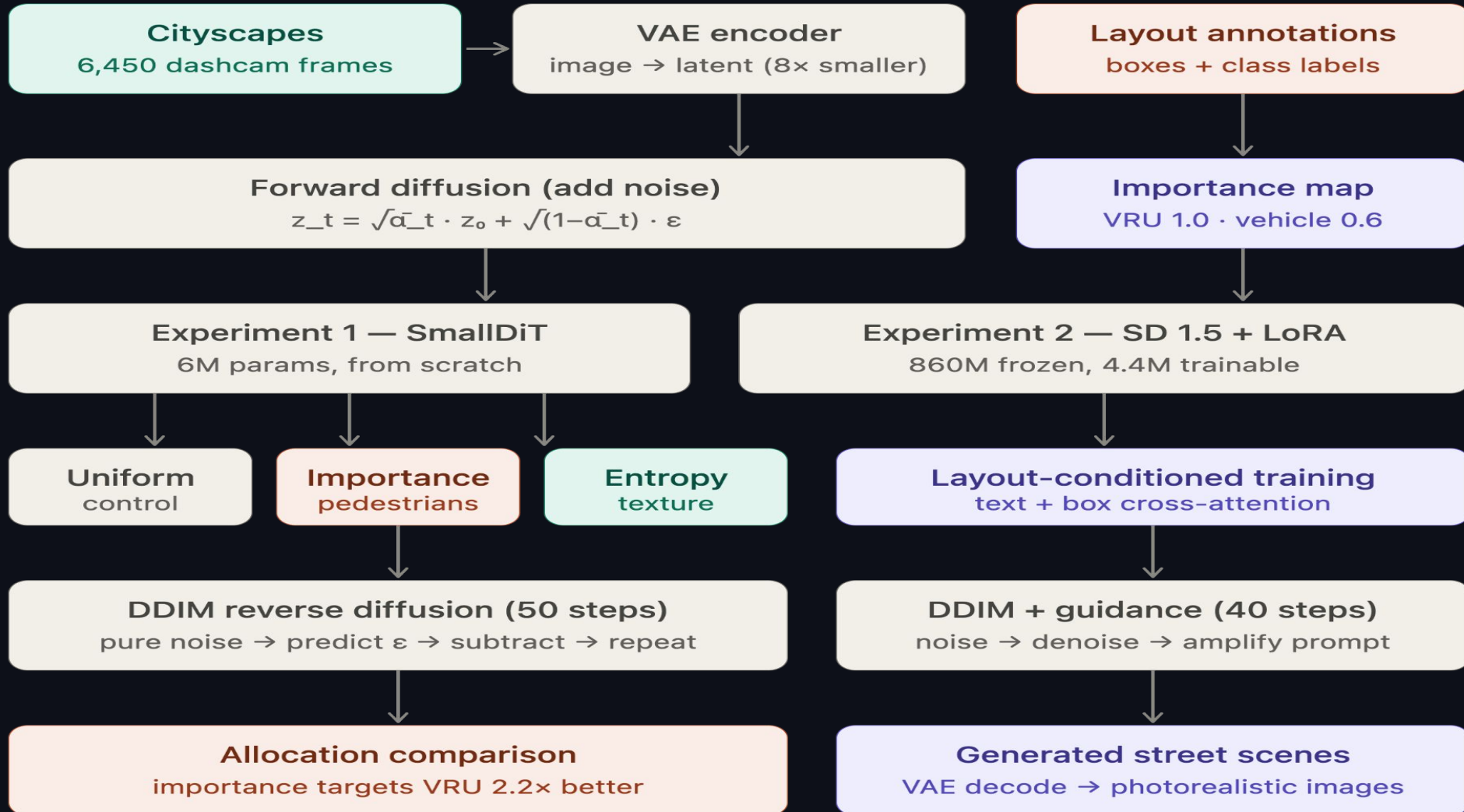
repeat millions of times with random images, timesteps and noise

Reverse process — generation

Start from pure noise and iteratively remove predicted noise

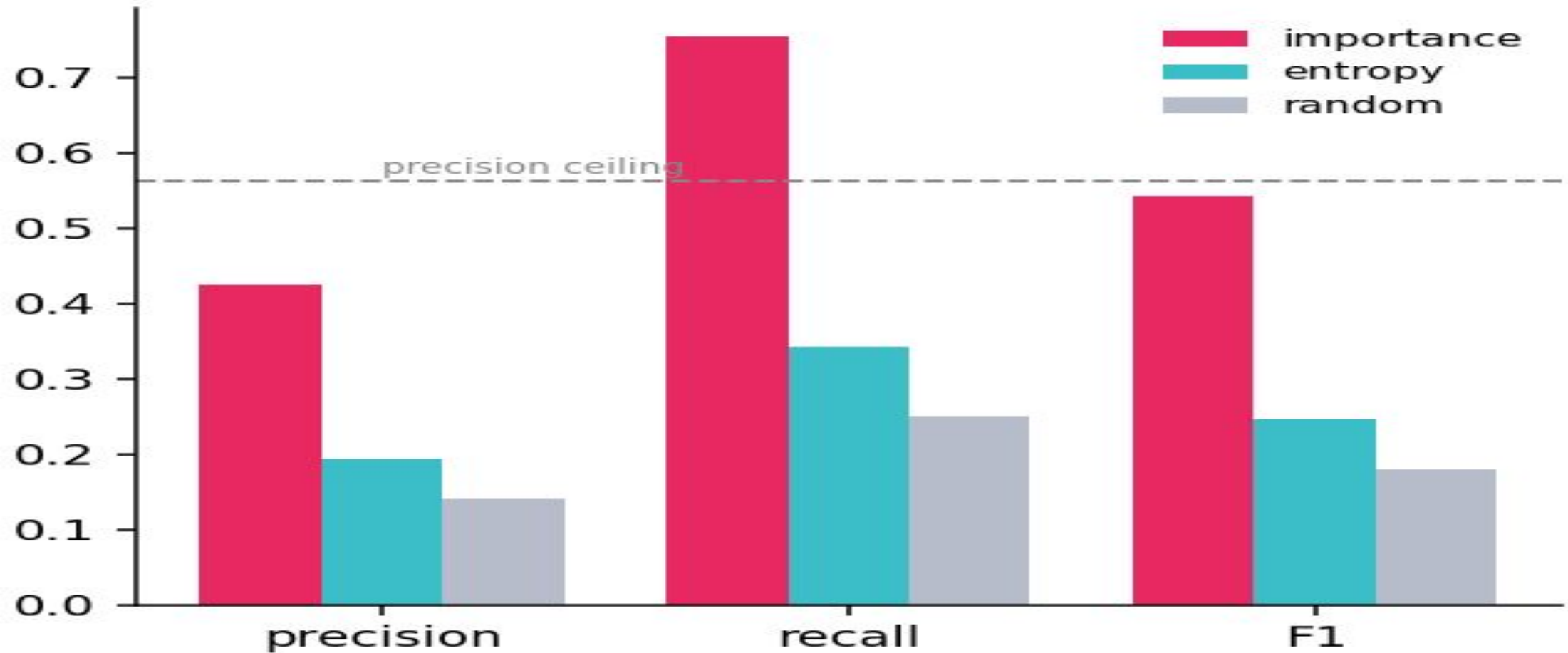


Architecture of two Experiments:



Importance reaches the pedestrian. Entropy does not.

Allocation as classification



Three-arm experiment: VRU error vs background error across training



GENERATION

LoRA fine-tuned SD 1.5 with 2,000 training steps



LoRA fine-tuned SD 1.5 with 6,000 training steps

6k steps val layouts seed 1 - guidance 7.5



Demo

COST

The whole study ran for the price of a sandwich

Stage	Wall clock (T4)	Cost
Data prep + grain-map precompute	0.3 h	\$0.16
Three-arm objective study (3 × 35 min)	1.8 h	\$0.95
Sampling + metrics + figures	0.5 h	\$0.27
LoRA fine-tune of SD 1.5	2.0 h	\$1.06
LoRA generation + validation grids	0.4 h	\$0.21
Core pipeline subtotal	5.0 h	\$2.65

NVIDIA T4, AWS g4dn.xlarge on-demand at \$0.526/h (Aug 2026). Spot drops to roughly \$0.20/h; T4 across providers runs \$0.35–0.59/h.

\$0.00

marginal cost of the method itself

The grain map is derived from layout, computed once, cached, and reused at every timestep. Swapping entropy for importance changes which cells get fine capacity, never how many.

Same tokens. Same FLOPs. Same wall clock.

COST AT SCALE

Generation is 50,000× cheaper than collection

Cost to synthesise 100,000 layout-conditioned frames · LoRA-adapted SD 1.5, 256×512, 40 DDIM steps

Hardware	Rate	Throughput [verify]	Wall clock	Total
T4	\$0.53/h	~360 img/h	~280 h	~\$147
A100 40 GB ← cost-optimal	\$1.10/h	~1,800 img/h	~56 h	~\$62
H100	\$2.50/h	~2,900 img/h	~34 h	~\$86

Throughput scales faster than price up to A100, then price outruns throughput. At 1 M frames the A100 figure is roughly \$620.

~\$30

per frame — Cityscapes-grade human annotation

Not a substitute: synthetic frames author tail cases collection cannot reach at any price.

~\$0.0006

per frame — this pipeline, on A100

Roughly a 50,000× ratio. Training from scratch at DiT-XL scale would be $O(10^3)$ GPU-hours — order \$3–10 k.

Key findings and technical outcomes

Semantic allocation decisively outperforms entropy-based allocation

- 01** Importance-weighted cell selection achieves 2.2× higher precision, recall and F1 over entropy. It captures 96.4% of the attainable precision ceiling versus entropy's 49%. Across 40/40 validation scenes, importance is favoured with $p = 1.8 \times 10^{-12}$. Entropy misses 2.7× more pedestrian cells, it wastes budget on cobblestones and tree textures.

LoRA fine-tuning enables photorealistic Cityscapes generation

- 02** Stable Diffusion 1.5 adapted with LoRA rank 16 (4.4M trainable parameters against 860M frozen) generates coherent, photorealistic German street scenes after just 2,000 steps (~2h on T4). At 6,000 steps the output quality improves further sharper pedestrians, better road markings, and more consistent dashcam perspective. The layout adapter successfully conditions object placement.

Larger models overfit, 6M parameters is the correct scale for this data

- 03** We also trained a 158.5M-parameter DiT-XL variant on the same 6,450 Cityscapes frames. At ~19 images per million parameters (versus DiT-XL/2's reference of ~1,896), the model memorised the training set rather than learning to generalise. The 6M SmallDiT at ~850 img/M-param trains stably and produces meaningful structure. This confirms: model capacity must match dataset scale.

What we would do next

01

Scale to larger datasets and higher resolution

Test on BDD100K dataset for domain transfer. Re-run at 512×1024 where 30% of pedestrians that currently occupy under half a latent cell would finally have resolution for fine tokens to act on.

02

Evaluate with downstream task metrics

Train a pedestrian detector on importance-allocated vs entropy-allocated synthetic data at equal frame count. Measure mAP on small pedestrians — a task metric, not a pixel metric.

03

Replace importance with learned risk fields

VRU 1.0 / vehicle 0.6 is a fixed. Replace with time-to-collision, occlusion state for per-scene adaptive allocation.

IN ONE SENTENCE

**Both signals get the same budget.
Entropy spends it on cobblestones.
Importance spends it on the cyclist.**

Only one of those is something a car needs to brake for.

